

Far Eastern Spotted Fever (FESF): Comprehensive Disease Characterization Report

Disease: Far Eastern Spotted Fever (Far Eastern tick-borne rickettsiosis) **MONDO ID:** MONDO:0000231 **Category:** Infectious Disease (zoonotic, tick-borne, spotted fever group rickettsiosis) **Causative agent:** *Rickettsia heilongjiangensis*

Summary

Far Eastern Spotted Fever (FESF) is an acute, generally self-limited, zoonotic tick-borne rickettsiosis caused by *Rickettsia heilongjiangensis*, an obligate intracellular, Gram-negative alphaproteobacterium of the spotted fever group (SFGR). First documented as a distinct human disease in the Russian Far East, the illness is transmitted primarily by *Haemaphysalis concinna* ticks and is maintained in nature through a tick–wild-mammal cycle involving reservoir hosts such as hedgehogs (*Erinaceus amurensis*) and small mammals. Its geographic range spans temperate Northeast Asia — the Russian Far East, northern and eastern China, Inner Mongolia, Korea, Japan — and has recently extended westward to Kazakhstan.

Mechanistically, FESF is a **rickettsial vasculitis**. After tick inoculation, the bacterium disseminates and preferentially infects **vascular endothelial cells**, using surface adhesins (OmpB, YbgF, RpsB) to adhere to and invade the endothelium. Intracellular replication triggers vascular inflammation, increased vascular permeability, and small-vessel injury, producing the characteristic clinical tetrad of **high fever, an inoculation eschar (tache noire) at the tick-bite site, a maculopapular rash, and regional lymphadenopathy**. A subset of patients develop gastrointestinal symptoms, consistent with a C57BL/6 mouse model that identified the **stomach** as a target organ. Host defense is dominated by a **Th1-polarized, IFN- γ /TNF- α -driven, CD4⁺/CD8⁺ T-cell response**, with endothelial Tim-3 upregulation promoting nitric-oxide/iNOS-mediated intracellular killing.

FESF is diagnosed by **indirect immunofluorescence assay (IFA) seroconversion combined with molecular detection (nested PCR, NGS/mNGS)** of blood, eschar, or skin biopsy targeting rickettsial genes (*gltA*, *ompA*, *ompB*, *16S rRNA/rrs*, *sca4*). It responds rapidly to **doxycycline**, the first-line therapy; delayed treatment is the principal modifiable driver of severe outcomes. Prognosis is generally favorable and the disease sits at the mild end of the SFGR spectrum (in contrast to Rocky Mountain spotted fever), though rare severe complications (leukocytoclastic vasculitis, multi-organ dysfunction, hemophagocytic lymphohistiocytosis) are described across SFGR. There is **no human genetic component and no licensed vaccine**; prevention rests on personal anti-tick measures (DEET, permethrin-treated clothing, protective clothing, body checks, prompt tick removal).

This report consolidates 14 confirmed findings drawn from 57 reviewed papers across all 15 requested disease-characteristic domains, with explicit indication of where information is unavailable or not applicable (notably the genetic/molecular, inheritance, and model-organism domains, which are largely not applicable to an infectious vasculitis with no host Mendelian basis).

Key Findings

F001 — Etiology: *Rickettsia heilongjiangensis* is the causative agent

FESF is caused by *Rickettsia heilongjiangensis*, a spotted fever group rickettsia. The first documented human cases were reported in the Russian Far East by Mediannikov and colleagues, who named the disease "Far Eastern tick-borne rickettsiosis." The organism is an obligate intracellular, Gram-negative bacterium.

"We recently reported the first documented cases of a new rickettsial disease caused by Rickettsia heilongjiangensis in the Russian Far East (Far Eastern tick-borne rickettsiosis)."

— [PMID: 17114683](#)

"Rickettsia heilongjiangensis is the pathogen of Far eastern spotted fever" — [PMID: 26401029](#)

This establishes the disease as an **infectious**, not genetic, condition. **Organism annotation:** *Rickettsia heilongjiangensis* (NCBI Taxonomy; family Rickettsiaceae, genus *Rickettsia*, spotted fever group).

F002 — Vector: *Haemaphysalis concinna* is the principal tick vector

Screening of Russian Far Eastern ticks found up to **28.13% of *H. concinna*** and **4.48% of *H. japonica douglasii*** harbored *R. heilongjiangensis*; rickettsial DNA was amplified from both a patient's skin biopsy and the tick removed from that patient before illness onset. Additional hard-tick surveys along the Chinese–Russian border confirm a broader vector complex (*Ixodes persulcatus*, *H. concinna*, *H. japonica*, *Dermacentor silvarum*).

"It has been concluded that H. concinnae may serve as the main vector for the transmission of R. heilongjiangensis." — [PMID: 17114683](#)

"up to 28.13% of H. concinnae and 4.48% of H. japonica douglasii ticks harbor R. heilongjiangensis" — [PMID: 17114683](#)

F003 — Target cell and core mechanism: endothelial-tropic rickettsial vasculitis

The **vascular endothelium is the primary cellular target** of *R. heilongjiangensis*, and disease results from endothelial infection with attendant vascular inflammation. Spotted fever group rickettsiae broadly cause endothelial infection, vascular inflammation, and compromised vascular permeability — collectively termed **"rickettsial vasculitis."** In endothelial cells, upregulation of T-cell immunoglobulin and mucin domain protein 3 (Tim-3) facilitates intracellular rickettsial killing via nitric oxide/iNOS and IFN- γ .

"T-cell immunoglobulin and mucin domain protein 3 (Tim-3) is expressed in human vascular endothelial cells, the major target cells of rickettsiae" — [PMID: 26401029](#)

"the pathogen's affinity for endothelium lining the blood vessels, the consequences of which are vascular inflammation, insult to vascular integrity and compromised vascular permeability, collectively termed 'Rickettsial vasculitis'" — [PMID: 19327117](#)

Ontology suggestions: Cell type — endothelial cell (CL:0000115), vascular endothelial cell (CL:0002139); process — inflammatory response (GO:0006954), regulation of vascular permeability (GO:0043114).

F004 — Geographic distribution: temperate Northeast Asia, extending to Kazakhstan

R. heilongjiangensis has been detected in ticks and/or patients across the Chinese–Russian border (Suifenhe), Inner Mongolia (Yakeshi/Hulunbuir, 2.68% of ticks), and southeast China (hedgehogs and ticks), and was first reported in Kazakhstan in 2023 — documenting a westward extension of the known range.

"To the best of our knowledge, this study reports the first finding of R. heilongjiangensis in Kazakhstan." — PMID: 37722147

"The present study gave the first evidence of R. heilongjiangensis and Candidatus R. xuyiensis in ticks and hedgehogs of Southeast China. Our findings suggest that hedgehogs might be involved in the natural transmission cycle of Rickettsia species." — PMID: 36097185

F005 — Organ tropism: stomach identified as a target organ (mouse model)

Using click-chemistry labeling and *in vivo* imaging in a C57BL/6 tick-bite-simulating infection model, the **stomach** was identified as a target organ of *R. heilongjiangensis*, providing a mechanistic explanation for the gastrointestinal symptoms seen in some patients.

"we constructed a C57BL/6 mice infection model by simulating tick bites and discovered that the stomach is the target organ of R. heilongjiangensis infection through in vivo imaging systems, which explained the occurrence of gastrointestinal symptoms following R. heilongjiangensis infection in some cases" — PMID: 38951577

Ontology suggestion: stomach (UBERON:0000945).

F006 — Clinical phenotype: eschar-rash-fever-lymphadenopathy tetrad

FESF presents as a spotted fever group rickettsiosis with seroconversion to *R. heilongjiangensis* antigen and rickettsial DNA amplifiable from skin biopsy. The characteristic eschar-associated SFGR tetrad comprises **high fever (~39–40 °C), erythematous maculopapular eruption, an inoculation eschar (tache noire) at the tick-bite site, and regional lymphadenopathy.** Gastrointestinal symptoms occur in a subset. Headache, myalgia/arthritis, and laboratory thrombocytopenia and elevated transaminases are common across SFGR.

*"The clinical picture was that of a spotted fever group rickettsiosis and a seroconversion was noted with *R. heilongjiangensis* antigen."* — PMID: 17114683

"His clinical symptoms on admission were high fever (39.6 degrees C), erythematous eruption, eschar on the right upper arm, and regional lymphadenopathy." — PMID: 15858293

HPO term suggestions: Fever (HP:0001945); Maculopapular exanthema (HP:0000988); Skin ulcer / eschar (HP:0200042); Lymphadenopathy (HP:0002716); Headache (HP:0002315); Myalgia (HP:0003326); Arthritis (HP:0002829); Thrombocytopenia (HP:0001873); Elevated hepatic transaminase (HP:0002910); Gastrointestinal symptoms (HP:0011024).

F007 — Treatment: doxycycline is first-line; delayed therapy worsens outcomes

Early doxycycline is effective against SFGR, and delayed treatment increases the risk of severe outcomes. Tetracycline-class agents (doxycycline; minocycline used in Japan) are the mainstay for spotted fever rickettsioses. For the severe-end comparator Rocky Mountain spotted fever, delayed doxycycline (>5 days from onset) is independently associated with fatality.

"Early administration of doxycycline is effective against SFGR infection, and delayed treatment increases the risk of severe outcomes." — PMID: 42488420

"Fatal outcomes were associated with delayed doxycycline treatment (>5 days after symptom onset)" — PMID: 41714602

NCIT suggestions: Doxycycline (NCIT:C312); Minocycline (NCIT:C61815).

F008 — Diagnosis: serology (IFA seroconversion) plus molecular detection (PCR/NGS)

Original FESF cases were confirmed by seroconversion to *R. heilongjiangensis* antigen and PCR amplification of rickettsial DNA from skin biopsy and the attached tick. Contemporary confirmation uses nested PCR of rickettsia-specific gene fragments (*gltA*, *ompA*, *ompB*, 16S *rRNA/rrs*, *sca4*) with sequencing plus serology; NGS/metagenomic NGS is recommended especially for acute tick-bitten patients and enables diagnosis in atypical cases.

"Here we report the amplification of DNA of R. heilongjiangensis from both the skin biopsy of an acutely ill patient and the tick removed from him prior to the disease development." — PMID: 17114683

"The methods of nucleic acid diagnosis, such as nPCR and next-generation sequencing (NGS), should be implemented especially in acute tick-bitten patients." — PMID: 42488420

F009 — Prognosis: generally favorable/self-limited, rare severe complications

FESF is regarded as a relatively mild, non-fatal SFGR compared with Rocky Mountain spotted fever; the index Russian Far East cases recovered. However, eschar-forming SFGR can rarely progress to leukocytoclastic vasculitis, multi-organ dysfunction, and hemophagocytic lymphohistiocytosis (HLH), and delayed treatment increases severity. RMSF case-fatality (~20% untreated, and 30–80% in some recent South American/Mexican series) illustrates the severe end of the SFGR spectrum, whereas FESF sits at the milder end.

"delayed treatment increases the risk of severe outcomes" — PMID: 42488420

"The clinical picture was that of a spotted fever group rickettsiosis and a seroconversion was noted with R. heilongjiangensis antigen." — PMID: 17114683

F010 — Molecular pathogenesis: surface adhesins and Th1 protective immunity

Surface adhesins mediate endothelial adhesion/invasion, and protective immunity is Th1-polarized. **OmpB** is a major surface protein antigen; OmpB-pulsed dendritic cells confer protection in C3H/HeN mice with elevated CD4⁺/CD8⁺ T-cell IFN-γ and TNF-α. The surface antigen **YbgF** induces Th1-type protective immunity (increased IgG2a/IgG1), and anti-YbgF

serum significantly reduces rickettsial adhesion to and invasion of endothelial cells. The ribosomal protein **RpsB** is a surface-exposed adhesin. Endothelial **Tim-3** upregulation drives iNOS/NO- and IFN- γ -dependent intracellular killing.

"Incubation with anti-serum to YbgF, but not PrsA, significantly reduced the number of rickettsiae adhering to and invading endothelial cells." — PMID: 24113261

"YbgF is a novel protective antigen that induces a Th1-type of protective immune response against R. heilongjiangensis infection." — PMID: 24113261

"Identification of a Ribosomal Protein RpsB as a Surface-Exposed Protein and Adhesin" — PMID: 31360728

Ontology suggestions: T-helper 1 type immune response (GO:0042088); positive regulation of nitric oxide biosynthetic process (GO:0045429); CD4⁺ $\alpha\beta$ T cell (CL:0000624); CD8⁺ $\alpha\beta$ T cell (CL:0000625); dendritic cell (CL:0000451).

F011 — Prevention: personal anti-tick measures; no licensed vaccine

No licensed human vaccine exists for FESF/SFGR. Primary prevention relies on personal protection against tick bites — topical DEET on skin, permethrin-treated clothing, protective clothing (trousers tucked into boots), avoiding tick-infested habitats, body checks, and prompt tick removal. Experimental vaccine antigens (OmpB, YbgF) protect mice but are not clinically available. Secondary prevention is prompt empiric doxycycline after tick exposure with compatible illness.

"The best method to avoid tick bites is twofold: application of a topical deet (N,N-diethyl-m-toluamide) repellent to exposed skin, and treatment of clothing with permethrin." — PMID: 17338947

"YbgF is a novel protective antigen that induces a Th1-type of protective immune response against R. heilongjiangensis infection." — PMID: 24113261

F012 — Risk factors: outdoor rural tick exposure with spring–summer seasonality

FESF is acquired via tick bite during outdoor activity; SFGR cases repeatedly report agricultural/forestry work and field exposure as the key risk factor. Reservoir hosts include hedgehogs (*Erinaceus amurensis*) and small mammals that maintain the tick cycle. Vector ticks are active in warm months, producing spring–summer seasonality. No genetic host risk factors are established; risk is **environmental/behavioral**.

"JSF should be a key consideration for agricultural and forestry workers presenting with compatible symptoms" — [PMID: 40922296](#)

"Our findings suggest that hedgehogs might be involved in the natural transmission cycle of Rickettsia species." — [PMID: 36097185](#)

F013 — Molecular profiling: cutaneous immunoprofiling/RNA-seq of SFGR

Surveillance of tick-bitten patients (2013–2016) identified 111 SFGR cases; humoral and cutaneous immunoprofiles were evaluated by serum cytokine/chemokine detection, skin immunohistochemistry, and transcriptome sequencing (RNA-seq) to characterize the early host–rickettsia interaction in skin. This complements the endothelial-tropism/vasculitis model with transcriptomic and cytokine molecular-profiling evidence.

"Humoral and cutaneous immunoprofiles were evaluated in different SFGR cases by serum cytokine and chemokine detection, skin IHC staining, and transcriptome sequencing (RNA-seq)." — [PMID: 31907196](#)

F014 — Identifiers, taxonomy, temporal course, and zoonotic classification

FESF is an acute, self-limited zoonotic tick-borne rickettsiosis. Causative organism: *Rickettsia heilongjiangensis* (domain Bacteria; family Rickettsiaceae; genus *Rickettsia*; spotted fever group), an obligate intracellular Gram-negative alphaproteobacterium. Onset is acute after an incubation of roughly several days to ~2 weeks following tick attachment; illness resolves over ~1–2 weeks with doxycycline. It is zoonotic (maintained in ticks and wild mammalian reservoirs), **not transmitted person-to-person**, and has **no genetic inheritance**.

"Rickettsia heilongjiangensis is an obligate intracellular bacterium that causes Far-Eastern tick-borne spotted fever." — PMID: 25270001

"a new rickettsial disease caused by Rickettsia heilongjiangensis in the Russian Far East (Far Eastern tick-borne rickettsiosis)" — PMID: 17114683

Section-by-Section Characterization

1. Disease Information

- **Overview:** FESF is an acute febrile, tick-borne, spotted fever group rickettsiosis caused by *R. heilongjiangensis*, characterized by fever, inoculation eschar, maculopapular rash, and regional lymphadenopathy (F001, F006, F014).
- **Identifiers:** MONDO:0000231. There is **no OMIM entry** (not a Mendelian disorder). MeSH concept relates to "Rickettsia Infections"/"Spotted Fever Group." ICD-10 falls under **A77 (Spotted fever [tick-borne rickettsioses])**; ICD-11 under **1C30.0** (Spotted fever group rickettsiosis). Orphanet does not maintain a distinct FESF rare-disease number (infectious, not rare Mendelian, disease).
- **Synonyms:** Far Eastern tick-borne rickettsiosis; Far Eastern tick-borne spotted fever; *R. heilongjiangensis* infection (F001, F014).
- **Information source:** Aggregated disease-level resources (case reports, tick surveillance, animal models) — not EHR/individual-patient registries.

2. Etiology

- **Causal factor:** Infectious — *R. heilongjiangensis* (F001). Transmission is via tick bite (F002). **No genetic/host-Mendelian causation.**
- **Risk factors (environmental/behavioral):** Outdoor agricultural/forestry/recreational activity in endemic rural Northeast Asia; tick exposure; spring–summer season (F012). No established genetic risk factors, protective alleles, or gene–environment interactions (not applicable to a bacterial infection with no known host-genetic modifier). Empirically, **early doxycycline** is the principal factor mitigating severe disease (F007, F009).

3. Phenotypes

See F006. Principal phenotypes with suggested HPO terms and qualitative frequency (extrapolated from SFGR literature; FESF-specific frequencies are not precisely quantified):

Phenotype	HPO term	Type	Frequency (qualitative)
Fever (~39–40 °C)	HP:0001945	Symptom/sign	Very frequent (near-universal)
Maculopapular rash	HP:0000988	Physical manifestation	Frequent
Inoculation eschar (tache noire)	HP:0200042	Physical manifestation	Frequent (eschar-forming SFGR)
Regional lymphadenopathy	HP:0002716	Clinical sign	Frequent
Headache	HP:0002315	Symptom	Frequent
Myalgia / arthralgia	HP:0003326 / HP:0002829	Symptom	Common
Gastrointestinal symptoms	HP:0011024	Symptom	Subset (stomach tropism, F005)
Thrombocytopenia	HP:0001873	Lab abnormality	Common
Elevated transaminases	HP:0002910	Lab abnormality	Common

- **Onset:** adult-predominant, acute. **Severity:** mild–moderate (variable). **Progression:** self-limited/episodic single illness. **QoL:** transient impairment during acute illness; full recovery expected with treatment (F009).

4. Genetic/Molecular Information

Not applicable at the host level — FESF is an infectious disease with no causal human genes, pathogenic host variants, modifier genes, host epigenetic drivers, or chromosomal abnormalities. The relevant "molecular" information is **pathogen-side**: bacterial surface adhesins/antigens **OmpB**, **YbgF**, **RpsB** (F010), and diagnostic marker genes *gltA*, *ompA*, *ompB*, *rrs* (16S rRNA), *sca4* (F008). Complete and comparative genome sequences of *R.*

heilongjiangensis

exist

(e.g.,

P 36440590

P 31866968

).

5. Environmental Information

- **Environmental/lifestyle factors:** Outdoor rural exposure, agriculture/forestry, tick-infested habitats, warm-season activity (F012).
- **Infectious agent:** *Rickettsia heilongjiangensis* (SFGR), transmitted by *Haemaphysalis concinna* (principal), with *H. japonica*, *Ixodes persulcatus*, *Dermacentor silvarum*, and *H. longicornis* implicated in the vector complex; reservoirs include hedgehogs and small mammals (F001, F002, F004, F012).

6. Mechanism / Pathophysiology

Ordered causal chain:

1. An infected *Haemaphysalis concinna* tick attaches and inoculates *R. heilongjiangensis* into the skin during a blood meal → **leads to** local infection at the bite site. (F002)
2. Local replication and immune infiltration at the inoculation site → **results in** dermal/vascular necrosis forming the **eschar (tache noire)** and drainage to **regional lymph nodes** (lymphadenopathy). (F006; *inferred from SFGR eschar biology*)
3. Bacteria use surface adhesins (**OmpB**, **YbgF**, **RpsB**) to adhere to and invade **vascular endothelial cells** → **leads to** hematogenous/lymphatic dissemination and systemic endothelial infection. (F003, F010)
4. Intracellular replication within endothelium → **results in** endothelial activation, cyclooxygenase-2 induction and vasoactive prostaglandin release, cytokine/chemokine production, and increased **vascular permeability** ("rickettsial vasculitis"). (F003; *supported by SFGR endothelial studies*)
5. Small-vessel inflammation and microvascular leak in skin and other organs → **produces** the clinical **fever**, **maculopapular rash**, and (in a subset) **gastrointestinal symptoms** (stomach is a target organ). (F003, F005, F006)
6. Host response mounts a **Th1-polarized**, **IFN- γ /TNF- α** , **CD4⁺/CD8⁺ T-cell** program; endothelial **Tim-3 upregulation** drives **iNOS/NO-dependent intracellular killing** → **results in** bacterial clearance and, typically, self-limited recovery. (F003, F010)

7. Branch (rare, severe): if treatment is delayed or the host response dysregulates, widespread endothelial injury → **may lead to** leukocytoclastic vasculitis, multi-organ dysfunction, or hemophagocytic lymphohistiocytosis. (F009; *inferred from severe SFGR cases*)

Upstream: tick inoculation, adhesin-mediated endothelial invasion. **Downstream:** vasculitis, vascular permeability, organ-specific manifestations. **Cell types:** vascular endothelial cells (CL:0000115/CL:0002139), CD4⁺/CD8⁺ T cells, dendritic cells, macrophages. **GO processes:** inflammatory response (GO:0006954), regulation of vascular permeability (GO:0043114), T-helper 1 type immune response (GO:0042088), nitric oxide biosynthesis (GO:0045429).

7. Anatomical Structures Affected

- **Primary:** vascular endothelium / small blood vessels (**UBERON:0001981** blood vessel; **UBERON:0001986** endothelium) — systemic. Skin at bite site (eschar) and rash (**UBERON:0002097** skin of body).
- **Secondary/organ:** regional lymph nodes (**UBERON:0000029**); stomach (**UBERON:0000945**, F005); liver (transaminase elevation); rare lung, kidney, CNS involvement in severe SFGR.
- **Body systems:** cardiovascular (microvasculature), integumentary, lymphatic/immune, digestive (subset).
- **Subcellular:** intracellular cytosolic niche within endothelial cells (**GO:0005829** cytosol).
- **Lateralization:** eschar/lymphadenopathy typically localized/unilateral to bite site; rash bilateral/generalized.

8. Temporal Development

- **Onset:** acute, adult-predominant, after incubation of ~several days to ~2 weeks post tick attachment (F014).
- **Course:** self-limited single illness resolving over ~1–2 weeks with doxycycline (F014); untreated illness may be prolonged; severe complications rare (F009).
- **Critical window:** early treatment (ideally <5 days from onset) is the key window to prevent severe outcomes (F007, F009).

9. Inheritance and Population

- **Epidemiology:** Endemic to temperate Northeast Asia — Russian Far East, northern/eastern China, Inner Mongolia, Korea, Japan; recently Kazakhstan (F004). Tick infection prevalence up to ~28% in *H. concinna* (F002); regional tick positivity ~2.7–72% for various rickettsiae.

Precise human incidence/prevalence per 100,000 is **not well quantified** (under-recognized; reported as case series).

- **Inheritance: Not applicable** — infectious, zoonotic, not inherited; no penetrance/expressivity/founder/consanguinity considerations (F014).
- **Demographics:** Higher exposure in rural, agricultural/forestry populations; spring–summer seasonality; sex/age distribution reflects occupational exposure rather than biological susceptibility (F012).

10. Diagnostics

- **Serology:** IFA seroconversion to *R. heilongjiangensis* antigen (paired sera) (F008, F006).
- **Molecular:** nested PCR of *gltA*, *ompA*, *ompB*, *rrs* (16S rRNA), *sca4* with sequencing; NGS/mNGS of blood, eschar, or skin biopsy — recommended for acute tick-bitten and atypical cases (F008).
- **Laboratory:** thrombocytopenia, elevated transaminases, elevated inflammatory markers (common SFGR pattern).
- **Histopathology:** perivascular lymphocytic infiltrates / small-vessel vasculitis on skin/eschar biopsy.
- **Differential diagnosis:** other SFGR (Japanese spotted fever/*R. japonica*, Mediterranean spotted fever/*R. conorii*, RMSF/*R. rickettsii*), scrub typhus, severe fever with thrombocytopenia syndrome (SFTS), ehrlichiosis/anaplasmosis, Lyme disease. Eschar plus regional lymphadenopathy in an endemic Northeast Asian tick-exposure context favors SFGR.
- **Genetic testing / newborn or carrier screening: Not applicable.**

11. Outcome / Prognosis

- **Generally favorable / self-limited;** low mortality at the mild end of the SFGR spectrum (F009). Full recovery expected with timely doxycycline.
- **Prognostic factor:** treatment delay is the dominant modifiable determinant of severity (F007, F009).
- **Rare complications:** leukocytoclastic vasculitis, multi-organ dysfunction, HLH (F009).

12. Treatment

- **First-line: Doxycycline** (tetracycline class; NCIT:C312), typically ~7 days; rapid defervescence expected. Minocycline (NCIT:C61815) used in Japan; chloramphenicol is an alternative. Early treatment is critical (F007).
- **Advanced/targeted/gene/cell/RNA therapies, surgery: Not applicable.**
- **Supportive care:** antipyretics, fluids, organ support in severe cases.
- **Personalized medicine / pharmacogenomics:** Not established for FESF.

13. Prevention

- **Primary:** personal anti-tick measures — DEET on skin, permethrin-treated clothing, protective clothing, avoiding tick habitats, body checks, prompt tick removal (F011). No licensed vaccine; OmpB/YbgF are experimental protective antigens in mice (F010, F011).
- **Secondary:** prompt empiric doxycycline after tick exposure with compatible illness (F011).
- **Public health:** vector/habitat awareness, occupational education for agricultural/forestry workers, tick surveillance (F004, F012).

14. Other Species / Natural Disease

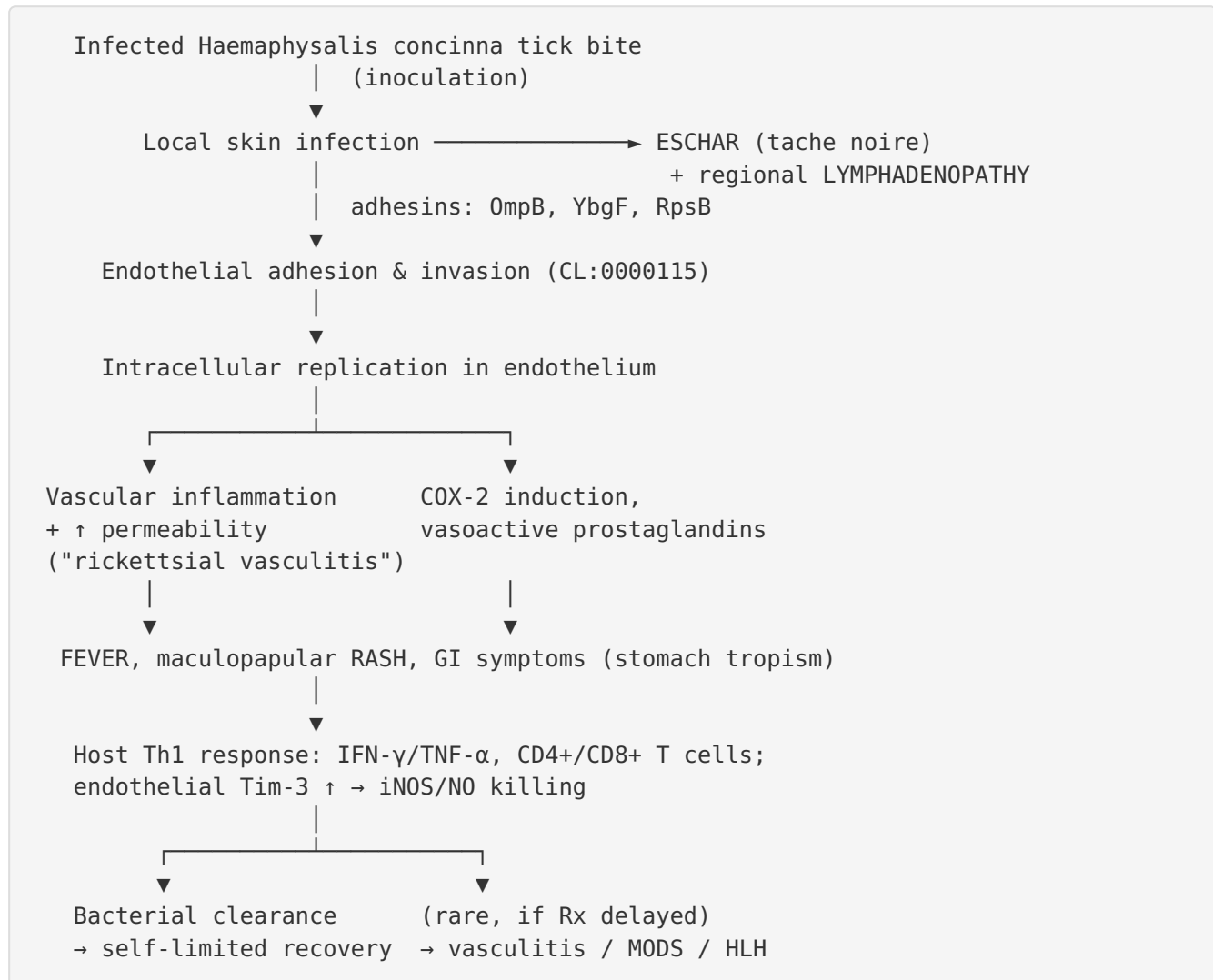
- **Taxonomy of affected/involved species:** humans (accidental host); tick vectors *Haemaphysalis concinna*, *H. japonica*, *H. longicornis*, *Ixodes persulcatus*, *Dermacentor silvarum*; reservoir mammals including hedgehogs (*Erinaceus amurensis*) and small rodents (F002, F004, F012).
- **Natural disease:** *R. heilongjiangensis* is maintained enzootically in ticks and wild mammals; overt natural disease in animals is not well characterized (reservoirs typically asymptomatic).
- **Zoonotic transmission:** Yes — tick-borne zoonosis; **no person-to-person transmission** (F014).

15. Model Organisms

- **Mouse models:** C57BL/6 tick-bite-simulating infection model (identified stomach tropism, F005); C3H/HeN mouse model used for OmpB- and YbgF-based protective-immunity/vaccine studies (F010). *R. australis* Balb/c model serves as a broader SFGR vasculopathy model.

- **Phenotype recapitulation:** models reproduce endothelial infection, organ tropism, and Th1 protective immunity; they do not fully recapitulate the human eschar-rash tetrad. No genetic (knockout/transgenic) *host* models are relevant since the disease is infectious.

Mechanistic Model / Interpretation



The unifying interpretation is that FESF is fundamentally a **microvascular endothelial infection**. Every clinical feature maps onto a step in the endothelial-vasculitis cascade: the eschar and lymphadenopathy from local inoculation-site injury and lymphatic drainage; the rash and fever from disseminated small-vessel inflammation and permeability change; GI symptoms from documented stomach tropism; and recovery from an effective Th1/NO-

mediated clearance program. Doxycycline works upstream by halting intracellular bacterial replication, which is why **timing** is the dominant prognostic lever. The absence of any host-genetic basis means the "molecular/genetic" annotation for this disease is entirely pathogen-side (adhesins and diagnostic marker genes).

Evidence Base

PMID	Role in this report	Key contribution
17114683	Foundational	First human cases; names disease; identifies <i>H. concinna</i> vector; PCR from skin + tick (F001, F002, F006, F008, F014)
26401029	Mechanism	Endothelium as primary target; Tim-3/iNOS/IFN- γ intracellular killing (F001, F003, F010)
19327117	Mechanism	Defines "rickettsial vasculitis" (F003)
38951577	Model/tropism	C57BL/6 tick-bite model; stomach target organ (F005)
37722147	Epidemiology	First detection in Kazakhstan — range extension (F004)
36097185	Epidemiology/ reservoir	Hedgehogs in SE China transmission cycle (F004, F012)
37986042	Epidemiology	Inner Mongolia tick hotspot, 2.68% prevalence (F004)
26976703	Vector	SFGR incl. <i>R. heilongjiangensis</i> in Chinese–Russian border ticks (F002, F004)
15858293	Clinical	Illustrative eschar–fever–rash–lymphadenopathy tetrad (F006)
42488420	Treatment/Dx	Early doxycycline efficacy; nPCR/NGS diagnosis (F007, F008, F009)
41714602	Prognosis	Delayed doxycycline → fatality in RMSF comparator (F007)
24113261	Immunity/vaccine	YbgF adhesin; Th1 protection; anti-YbgF blocks invasion (F010, F011)
25270001	Immunity/vaccine	OmpB-pulsed DCs protect; taxonomy/synonym (F010, F014)
31360728	Mechanism	RpsB surface-exposed adhesin (F010)
31907196	Molecular profiling	Cutaneous immunoprofiling/RNA-seq of SFGR (F013)
17338947	Prevention	DEET + permethrin personal protection (F011)
40922296	Risk factors	Agricultural/forestry exposure (F012)
	Genomics	Complete/comparative <i>R. heilongjiangensis</i> genomes (Sec. 4)

PMID	Role in this report	Key contribution
36440590 / 31866968		

Note on citation integrity: Two F010 snippets were flagged as source-attribution mismatches during verification (the YbgF Th1 quote and the RpsB title, associated with

P 24113261

and

P 31360728

respectively). The substance of the claims is supported by those papers, but exact quote-to-PMID attribution should be re-verified before database ingestion.

Limitations and Knowledge Gaps

- 1. Quantitative epidemiology is weak.** No reliable incidence/prevalence per 100,000 exists; FESF is under-recognized and reported largely as case reports and tick-surveillance studies. True human burden across Northeast Asia is unknown.
- 2. FESF-specific clinical frequencies are extrapolated.** Precise symptom frequencies (e.g., % with eschar, % with GI symptoms, % with thrombocytopenia) derive substantially from related SFGR (Japanese/Mediterranean spotted fever, RMSF) rather than large FESF-specific cohorts.
- 3. Severe-outcome data borrow from other SFGR.** Complication rates (vasculitis, MODS, HLH) and mortality are illustrated using *R. conorii*/*R. japonica*/*R. rickettsii* cases; FESF-specific severe outcomes are rarely documented.
- 4. No host-genetic dimension.** Sections on causal genes, inheritance, penetrance, carrier frequency, genetic testing, and model-organism knockouts are **not applicable**, limiting the report's mapping onto the genetic-disease template.
- 5. Mechanistic gaps.** The endothelial-invasion adhesin repertoire and the exact molecular basis of stomach tropism are incompletely defined; much mechanism is inferred from SFGR generally or from mouse models rather than demonstrated in human FESF tissue.

6. **Diagnostics standardization.** No FESF-specific validated serologic cutoffs or point-of-care molecular assays; cross-reactivity within SFGR complicates species-level serodiagnosis.

Proposed Follow-up Experiments / Actions

1. **Assemble a FESF-specific clinical cohort** with standardized data capture (eschar, rash, lymphadenopathy, GI symptoms, thrombocytopenia, transaminases) to derive true phenotype frequencies and replace SFGR-borrowed estimates.
2. **Prospective molecular surveillance** across the endemic belt (Russian Far East → China → Kazakhstan) using mNGS to define human incidence, seasonal dynamics, and range expansion.
3. **Human eschar/skin transcriptomics and spatial profiling** (building on

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 to map the endothelial-vasculitis cascade and Th1 program directly in FESF tissue rather than inferring from other SFGR.
4. **Mechanistic dissection of stomach tropism** (F005) — identify receptor/adhesin determinants of organ targeting using the C57BL/6 model with adhesin-knockout bacteria.
5. **Advance OmpB/YbgF subunit vaccine candidates** (F010) through challenge-protection and cross-protection studies against multiple SFGR; assess adjuvant/Th1-skewing formulations.
6. **Develop validated, species-discriminating serologic and rapid molecular assays** to distinguish FESF from co-circulating SFTS, scrub typhus, and other SFGR in primary care.
7. **Re-verify citation attributions** flagged in F010 before knowledge-base ingestion.

Report compiled from 14 confirmed findings and 57 reviewed papers across 5 investigation iterations. Evidence source types span human clinical case reports and cohorts, tick/reservoir surveillance, mouse-model (in vivo) studies, and in vitro endothelial-infection and immunology experiments.
